

EXPERIMENT 4

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D20A-02

AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY:

1. MetaMask

MetaMask is a **cryptocurrency wallet and browser extension** that allows users to interact with blockchain networks and decentralized applications (DApps). It manages the user's blockchain accounts, including their public wallet address and private keys, and allows users to approve or reject blockchain transactions. In this experiment, MetaMask acts as the user's wallet for connecting to a blockchain test network and signing transactions generated through Remix IDE.

MetaMask can be connected to Remix IDE so that a Solidity smart contract deployed from Remix uses the account selected in MetaMask. When a transaction is initiated, MetaMask displays the transaction details, such as the destination address, gas fee, and network, and asks the user to confirm the transaction.

2. Testnet

A **testnet** is a blockchain network designed for development and testing purposes. It provides an environment that behaves similarly to a real blockchain network but uses **test cryptocurrency**, which has no real monetary value. Developers can use testnets to deploy and test smart contracts and perform transactions without risking real funds.

In this experiment, the **Sepolia Testnet** or **RSK Testnet** can be used. Test ETH on Sepolia or test RBTC on RSK is used to pay transaction fees during testing. The use of a testnet allows the smart contract to be compiled, deployed, and interacted with safely before considering deployment on a production blockchain network.

3. Remix IDE

Remix IDE is a browser-based development environment used for writing, compiling, deploying, and interacting with Ethereum-compatible smart contracts written in Solidity. It provides a Solidity code editor, compiler, deployment interface, and tools for interacting with deployed contracts.

For this experiment, Remix IDE is connected to MetaMask through the **Injected Provider - MetaMask** environment. This allows transactions initiated from Remix to be signed using the selected MetaMask account.

4. Solidity Smart Contracts

A **smart contract** is a program deployed on a blockchain that automatically executes predefined logic when its functions are called. Solidity is a programming language commonly used to write smart contracts for Ethereum-compatible blockchain networks.

In this experiment, a simple Solidity smart contract based on the selected mini-project is created. The contract is first compiled in Remix IDE and then deployed to the selected testnet through the MetaMask wallet.

5. Connecting MetaMask with Remix IDE

The general process for connecting MetaMask with Remix IDE is:

1. Install the MetaMask browser extension.
2. Create a new wallet or import an existing account.
3. Select the required test network, such as **Sepolia**.
4. Obtain test cryptocurrency from a testnet faucet to pay transaction fees.
5. Open Remix IDE and create the Solidity smart contract.
6. Compile the smart contract using the Solidity Compiler.
7. Open **Deploy & Run Transactions** in Remix.
8. Select **Injected Provider - MetaMask** as the environment.
9. MetaMask will request permission to connect the selected account to Remix.
10. Approve the connection.
11. Deploy the compiled smart contract.
12. Confirm the transaction through MetaMask.
13. After the transaction is confirmed, interact with the deployed smart contract through Remix.

6. Transactions and Gas

A **transaction** is an operation submitted to the blockchain that changes its state, such as deploying a smart contract or modifying data stored by a smart contract. Transactions must be signed by the user's blockchain account.

Gas represents the computational work required to execute an operation on the blockchain. A transaction requires a certain amount of gas, and the user pays a transaction fee using the native cryptocurrency of the selected network.

When a transaction is initiated from Remix, MetaMask displays the transaction and its associated fee. The user can review the details and confirm the transaction. Once confirmed and included in a block, the transaction receives a transaction hash that can be used to view its status and details on a blockchain explorer.

7. Blockchain Explorer

A **blockchain explorer** is a web-based tool used to inspect blockchain activity. It allows users to search for and view information about transactions, blocks, wallet addresses, and smart contracts.

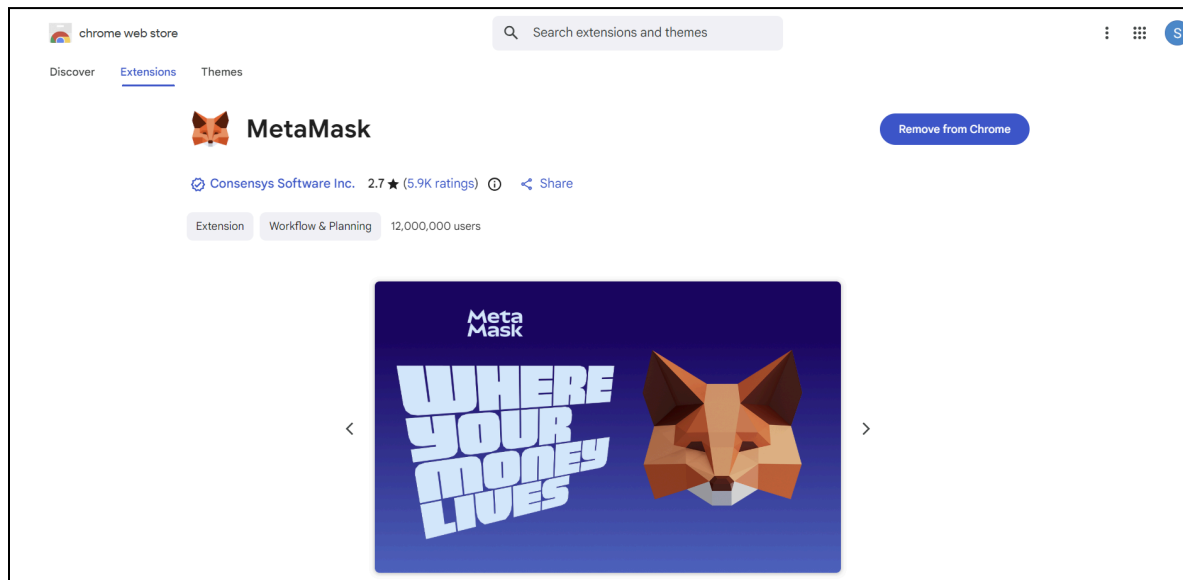
After deploying and interacting with the smart contract, the transaction hash can be viewed on the appropriate testnet explorer. The explorer can show details such as:

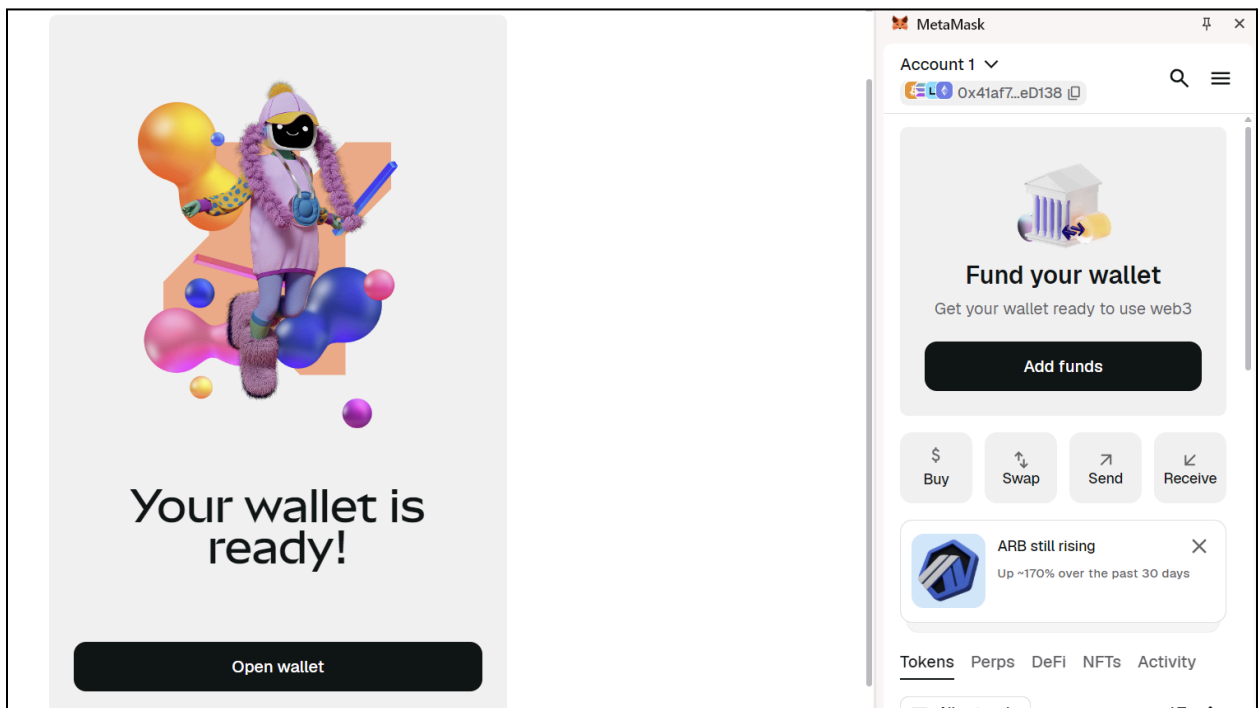
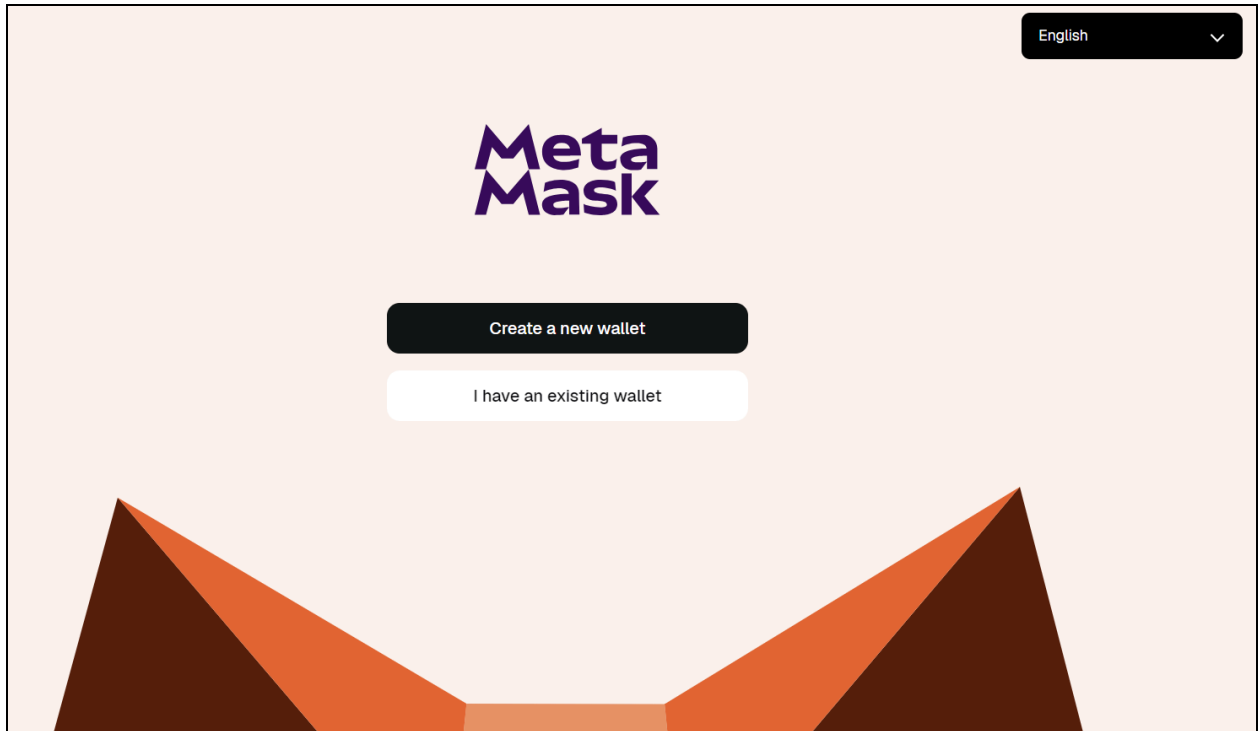
- Transaction hash
- Sender address
- Receiver/contract address
- Transaction status
- Block number
- Gas used
- Transaction fee
- Timestamp

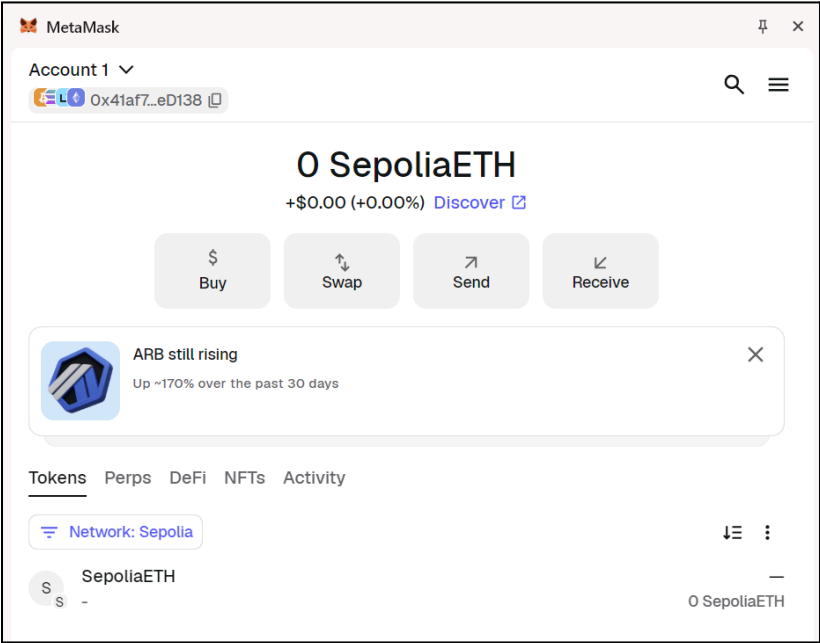
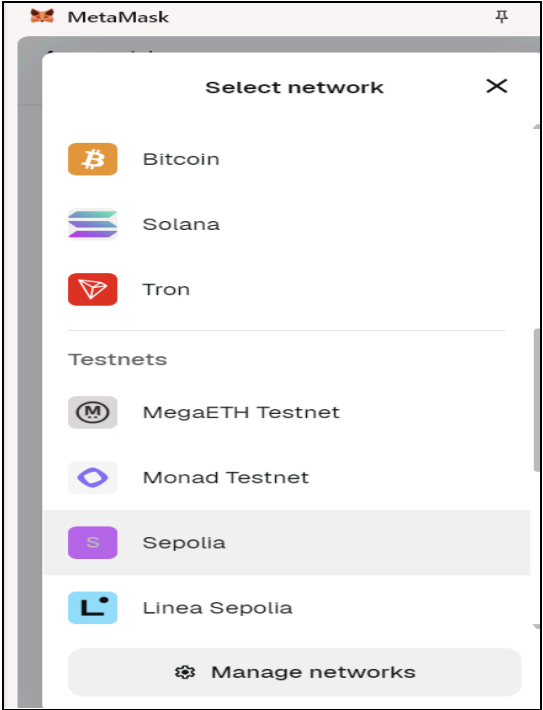
8. Interaction with the Smart Contract

After successful deployment, Remix displays the deployed contract and its available functions. Functions that modify blockchain data require transactions and therefore require confirmation through MetaMask. Read-only functions can generally be called without submitting a state-changing transaction.

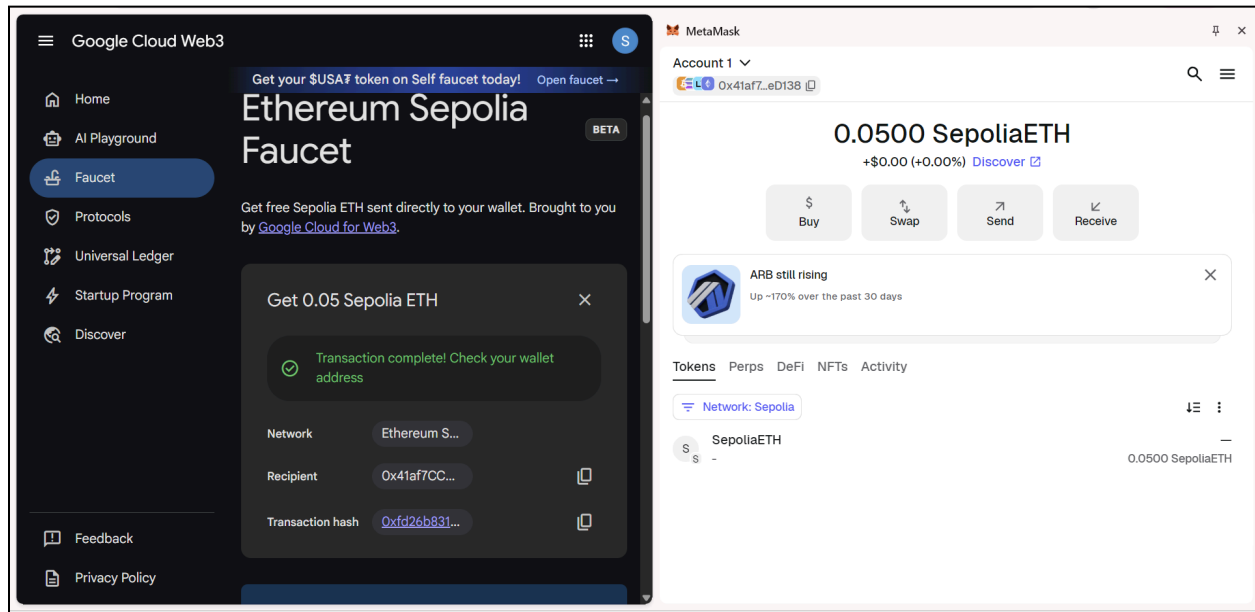
CODE AND OUTPUT:



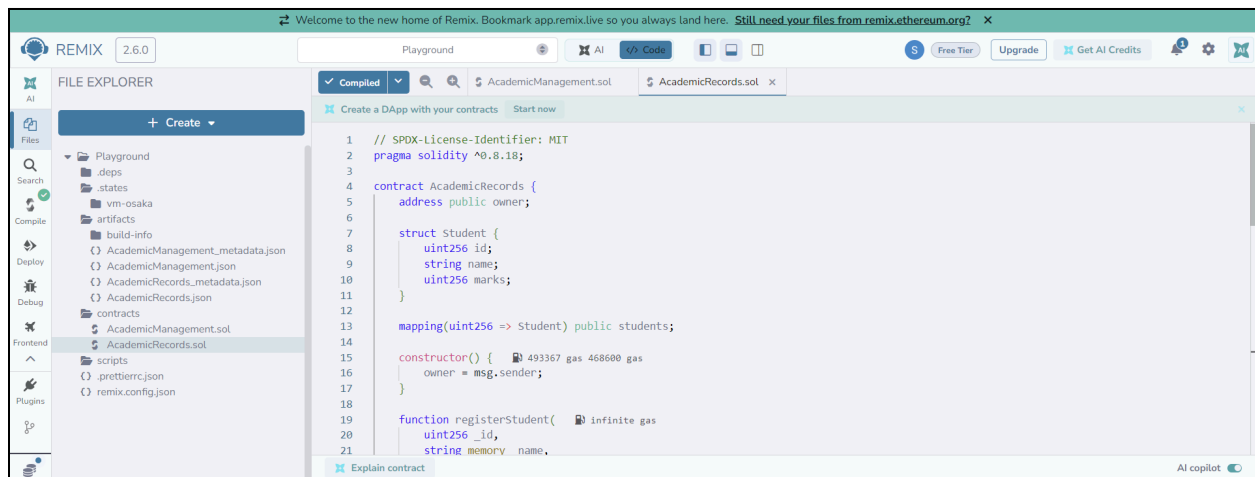




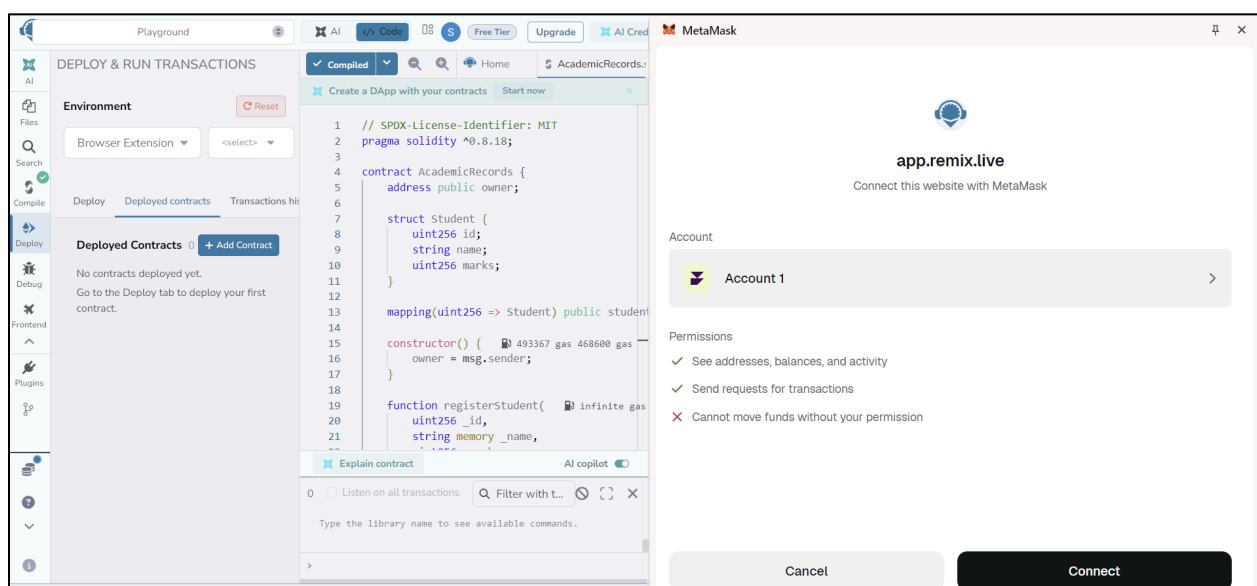
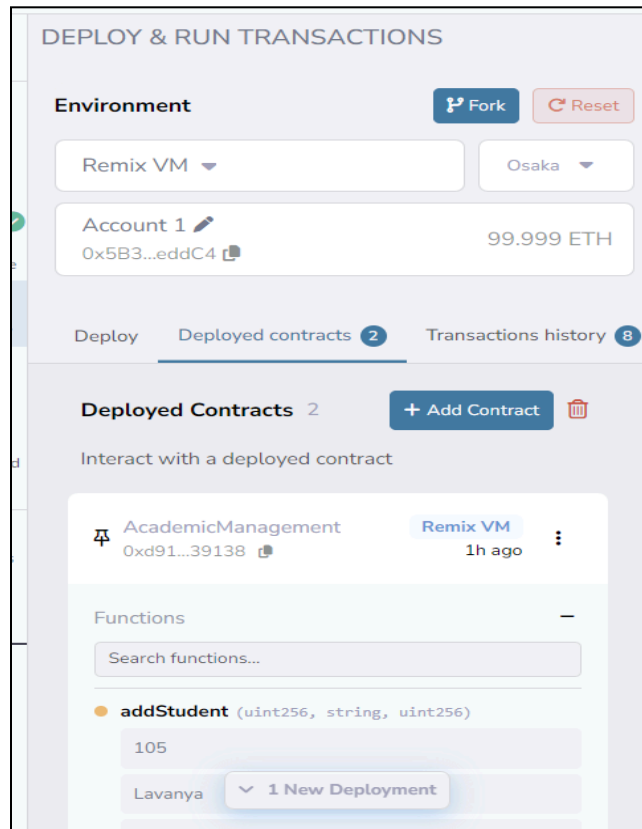
GOOGLE CLOUD FAUCET:



Academic Record Management code:



Deploying using MetaMask:



DEPLOY & RUN TRANSACTIONS

Environment

Browser Extension

Sepolia Testnet - M

Account 16
0x41a...ed138

0.050 ETH

Reset

DeployDeployed contractsTransactions his

Deploy

Sepolia (11155111)

AcademicRecords
AcademicRecords.sol

✓ Compiled

Verify Contract on Explorers

Value0wei

Gas limitauto0

Deploy

DEPLOY & RUN TRANSACTIONS

Environment

Browser Extension

Sepolia Testnet - MetaMask

Account 16
0x41a...ed138

0.050 ETH

Reset

DeployDeployed contractsTransactions history

Deploy

Sepolia (11155111)

AcademicRecords
AcademicRecords.sol

✓ Compiled

Verify Contract on Explorers

Value0wei

Gas limitauto0

Deploy

✓ Compiled

MetaMask

1 of 2

Reject all

Account 1

Deploy a contract

This site wants you to deploy a contract

Estimated changes ⓘNo changes

NetworkSepolia

Request from ⓘapp.remix.live

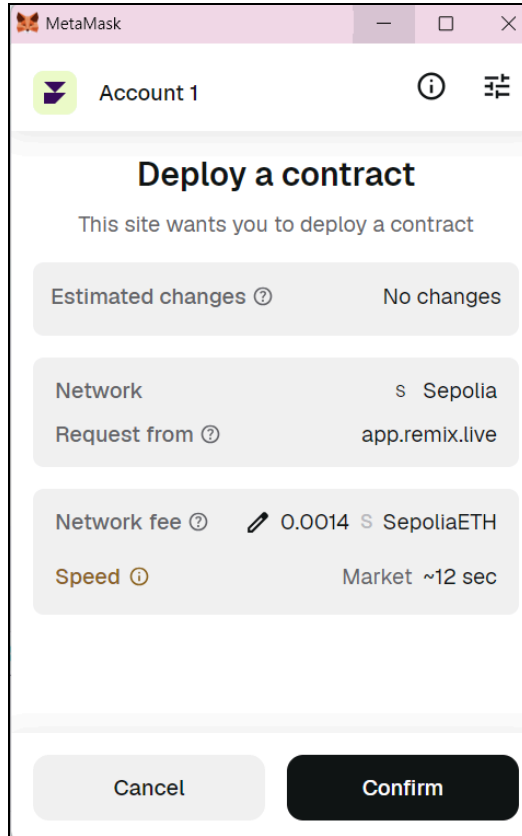
Network fee ⓘ0.0015 SepoliaETH

SpeedMarket ~12 sec

Cancel

Confirm

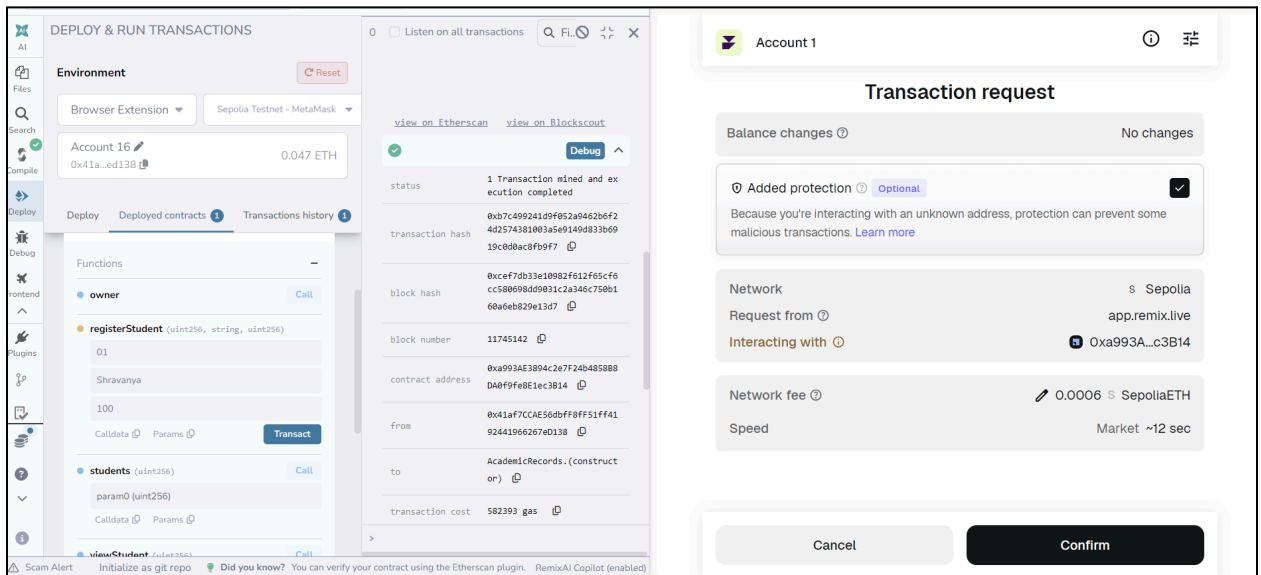
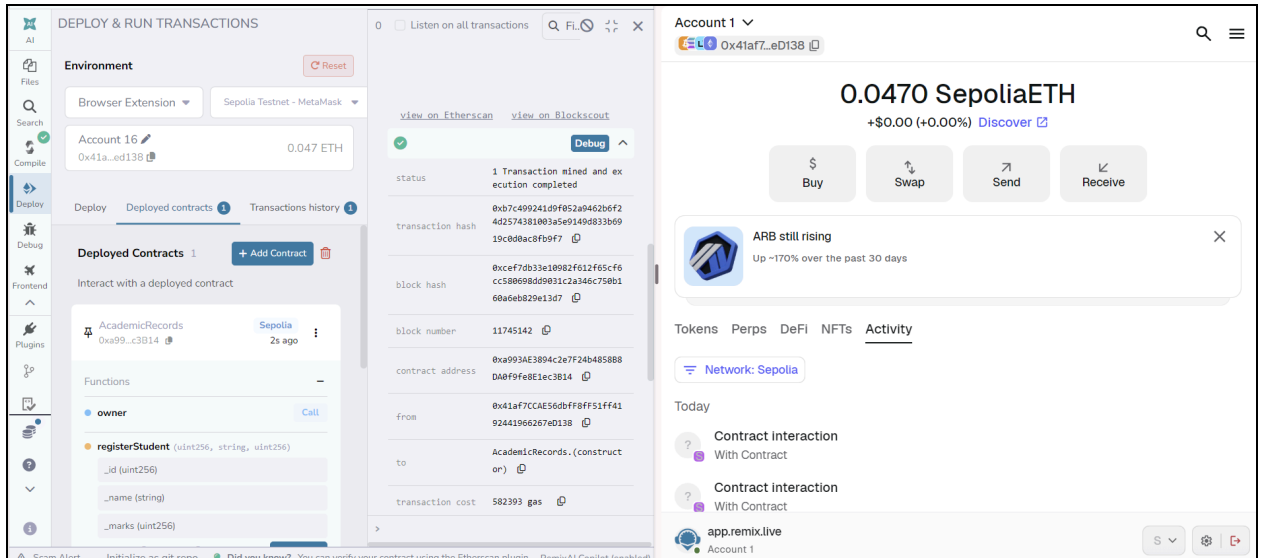
am AlertInitialize as git repoDid you know? You can verify your contract using the Eth



✓

Debug ^

status	1 Transaction mined and execution completed	
transaction hash	0xb7c499241d9f052a9462b6f24d2574381003a5e9149d833b6919c0d0ac8fb9f7	🔗
block hash	0xcfe7db33e10982f612f65cf6cc580698dd9031c2a346c750b160a6eb829e13d7	🔗 Copy
block number	11745142	🔗
contract address	0xa993AE3894c2e7F24b485888DA0f9fe8E1ec3B14	🔗
from	0x41af7CCA56dbFF8FF51ff4192441966267eD138	🔗
to	AcademicRecords.(constructor) 🔗	
transaction cost	582393 gas	🔗
decoded input	{ } 🔗	
decoded output	- 🔗	
logs	[] 🔗	
raw logs	[] 🔗	



Register student:

[illegible]

View Student:

[illegible]

[illegible]

Registration hash verified:

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

 Etherscan

HomeBlockchain ▾Tokens ▾NFTs ▾API ▾More ▾

Transaction Details< >

Sponsored:  Etherscan API - Onchain data for your agents across 60+ chains via MCP, CLI, Skills. [Build with AI.](#)

OverviewLogs (1)AuthorizationsState

</> API

 TRANSACTION ACTION

EIP-7702: 0x41af7CCA...6267eD138 Delegate to 0x63c0c19a...A07DAE32B

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0x8c9f31e46173be7919666c517738ae47f058f1e21c4889e853c40dcf8002ef3b

Status:

Success

Block:

1174519816 Block Confirmations

Timestamp:

3 mins agoSep-20-2026 03:50:00 PM +UTC

From:

0x41af7CCA56dbf8f51ff4192441966267eD138

To:

0xdb9B1e94B5b69Df7e401DDbedE43491141047dB3

Transaction Hash:	0x8c9f31e46173be7919666c517738ae47f058f1e21c4889e853c40dcf8002ef3b
Status:	Success
Block:	11745198 16 Block Confirmations
Timestamp:	3 mins ago Sep-20-2026 03:50:00 PM +UTC
From:	0x41af7CCAE56dbf8f51ff4192441966267eD138
To:	0xdb9B1e94B5b69Df7e401DDbedE43491141047dB3
Value:	0 ETH
Transaction Fee:	0.000481009101349332 ETH
Gas Price:	2.497814331 Gwei (0.000000002497814331 ETH)
Gas Limit & Usage by Txn:	226,291 192,572 (85.1%)
Gas Fees:	Base: 0.997814331 Gwei
Burnt Fees:	Burnt: 0.000192151101349332 ETH (\$0.00)
Other Attributes:	Txn Type: 4 (EIP-7702) Nonce: 2 Position In Block: 8
Input Data:	Function: redeemDelegations(bytes[] _permissionContexts,bytes32[] _modes,bytes[] _executionCallDatas)

Deployment hash verified:

TRANSACTION ACTION
Contract Created at 0xa993AE38...8E1ec3B14

[This is a Sepolia Testnet transaction only]

Transaction Hash:	0xb7c499241d9f052a9462b6f24d2574381003a5e9149d833b6919c0d0ac8fb9f7
Status:	Success
Block:	11745142 76 Block Confirmations
Timestamp:	16 mins ago Sep-20-2026 03:38:00 PM +UTC
From:	0x41af7CCAE56dbf8f51ff4192441966267eD138
To:	0xa993ae3894c2e7f24b4858b8da0f9fe8e1ec3b14 Created
Value:	0 ETH
Transaction Fee:	0.001459572908000739 ETH
Gas Price:	2.506164923 Gwei (0.000000002506164923 ETH)

More Details:
+ Click to show more

A transaction is a cryptographically signed instruction that changes the blockchain state. Block explorers track the details of all transactions in the network. Learn more about transactions in our Knowledge Base.

Deployed Contract Address verified:

Etherscan

HomeBlockchainTokensNFTsAPIMore

Contract

0xa993AE3894c2e7F24b4858B8DA0f9fe8E1ec3B14

Sponsored:

Etherscan API - Onchain data for your agents across 60+ chains via MCP, CLI, Skills. Build with AI.

Overview

ETH BALANCE
0 ETH

More Info

CONTRACT CREATOR
0x41af7CCA...6267eD138 | 19 mins ago

Multichain Info

N/A

Transactions

Token Transfers (ERC-20)

Contract

Events

Transaction Hash

Method

Block

Age

From

To

Amount

Txn Fee

METAMASK Balance:

MetaMask

Account 1

0x41af7...eD138

0.0466 SepoliaETH

+\$0.00 (+0.00%) Discover

Buy

Swap

Send

Receive

ARB still rising

Up ~170% over the past 30 days

Tokens

Perps

DeFi

NFTs

Activity

Network: Sepolia

Today

Contract interaction
With 0xdb9b1...47db3

Contract interaction
With Contract

Contract interaction
With Contract

CONCLUSION:

The experiment successfully demonstrated the development and deployment of a Solidity-based blockchain application using Remix IDE and MetaMask. The **AcademicRecords** smart contract was compiled and deployed on the Sepolia Testnet, and student information was successfully stored and retrieved through blockchain transactions. MetaMask was used to approve and execute transactions, while Sepolia Etherscan was used to verify the transaction details. The experiment provided practical understanding of smart contract deployment, wallet integration, blockchain transactions, gas consumption and interaction with deployed contracts.